Pre-FEED design gate

DRAFT — simulator screening, not approved for construction.

Verdict: NOT ROBUST across 4 stated operating scenarios.

Operating envelope

Scenario	Load	Result	Governing check	Utilisation	Mass closure	Basis
Clean design at today's rate	100%	PASS	V-201 boil-up duty vs heat from the heating steam (2 % closure tolerance)	0.98	0%	US 5,366,514 Table I operating point; E-201 clean U is the illustrative post-clean value in basis.md
Clean design at sales target	120%	PASS	K-201 suction flow vs capacity	1	0%	basis.md request: 3,000 short t/d (+20%)
Target with heater fouled back to current U	120%	FAIL	V-201 heater area needed (BPE included) vs installed	1.11	0%	basis.md equipment ratings: E-201 U 2,000 W/m2K in service
Target with five-point existing-compressor efficiency sensitivity	120%	PASS	K-201 suction flow vs capacity	1	0%	engineering sensitivity around the provisional 75% efficiency; vendor curve required

Energy closure: not available: boundary streams do not carry enthalpy_kJ_kg

Preliminary equipment selection

Tag	Type	Status	Flow margin	Driver margin	Rating source
K-202	compressor	provisional pass	—	1.19	vendor quote to confirm / standard size 55 t/h suction, 2,400 kW shaft at pressure ratio 1.8 (3.08 bar discharge) - proposed rating only, not an existing rating

Vendor gate still open: guaranteed curve; efficiency/driver; surge/choke or BEP/minimum flow; materials; vibration/noise; mechanical design.

Preliminary equipment datasheet

Provisional equipment selection only. This is not a vendor datasheet and not a mechanical design approval.

K-202 compressor

- Service: compressor
- Inlet: S3001_K202, 115.15 °C, 1.7 bar, phase vapour
- Outlet: S3002_K202, 191.5 °C, 3.08 bar, phase vapour
- Operating flow: to confirm
- Pressure ratio: to confirm; head: to confirm
- Calculated duty: 2.02068 MW
- Assumed efficiency: to confirm %
- Provisional rating: 55 t/h / 2.4 MW
- Rating source: vendor quote to confirm / standard size 55 t/h suction, 2,400 kW shaft at pressure ratio 1.8 (3.08 bar discharge) - proposed rating only, not an existing rating
- Missing vendor information: vendor curve, mechanical design, materials — to confirm
- Status: new equipment, rating to confirm

Critical-loop dynamics

Not run: no measured dynamic model.

Needed: equipment liquid/vapour holdup; controller action and tuning; valve/drive response; step-test time constant and dead time; trip and permissive logic.

Change-focused HAZOP and relief screen

Tag	Deviation	Cause	Consequence	Existing safeguard	Action
V-201	High pressure	vapour outlet blocked	vessel overpressure	protected	verify relief load and discharge destination
V-201	High level	outlet restriction/ control failure	carryover to compressor	protected	confirm independent high-high trip

K-202	No flow / trip	trip, isolation or surge	loss of production; recycle heating	new equipment: relief and isolation to confirm	confirm anti-surge, trip and restart logic
K-202	High pressure	blocked discharge or control failure	overpressure upstream/downstream	new equipment: relief and isolation to confirm	size relief/depressuring case; check discharge system
K-201	No flow / trip	trip, isolation or surge	loss of production; recycle heating	protected	confirm anti-surge, trip and restart logic
K-201	High pressure	blocked discharge or control failure	overpressure upstream/downstream	protected	size relief/depressuring case; check discharge system
D-201MIX	No flow	isolation or failure	loss of process function	not shown on relief scan	confirm alarms, trips and operating response
OVERHEAD_SPLIT	No flow	isolation or failure	loss of process function	not shown on relief scan	confirm alarms, trips and operating response

Cost and constructability gate

Cost classes on the proposals: <$100k, >$1M, no capex.

Outage required: yes. Exact tie-ins: to confirm on the Rev B line list.

No sourced numerical estimate supplied; these items remain unpriced: installed equipment; piping/tie-ins; electrical and controls; civil/structural; outage production loss.

- Plot space, access and lifting study
- Foundation and nozzle-load design
- Electrical load, starter and cable study
- Vendor lead time and spares
- Materials/corrosion review
- Tie-in list, isolation plan and shutdown window
- Line sizing, supports, stress and hydrotest
- Control narrative and cause/effect update
- Cleaning method, waste handling, inspection and post-clean performance test

Gate and limitations

- A pass means all stated simulator checks passed in all stated scenarios; it is not mechanical or vendor approval.
- New equipment remains provisional until guaranteed curves and materials are checked.
- The automated hazard register does not replace a multidisciplinary HAZOP or relief-system design.
- Missing inputs stay open; PlantAgent does not invent curves, dynamic constants or costs.